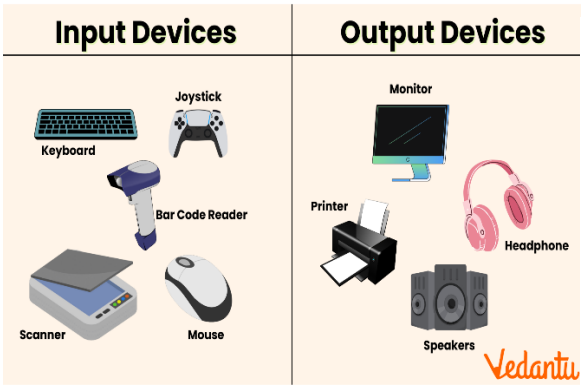
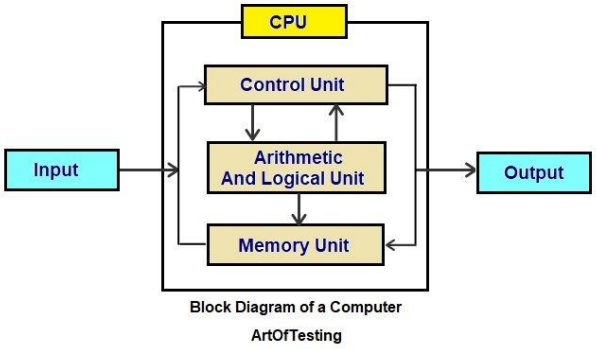
Assembling and Configuration of Computer Hardware

Introduction to Computer Hardware





1.1 Computer System Overview

A computer system is an electronic device that accepts data (input), processes it according to a set of instructions, stores it, and produces meaningful results (output).

A computer system is a combination of **hardware**, **software**, **data**, **users**, and **procedures** that work together to perform computing tasks efficiently and accurately.

Major components of a computer system:

- **Input Unit** – Keyboard, mouse, scanner
- **Processing Unit** – CPU
- **Memory Unit** – RAM, ROM
- **Storage Unit** – HDD, SSD
- **Output Unit** – Monitor, printer

Input Unit → CPU (Processing) → Output Unit

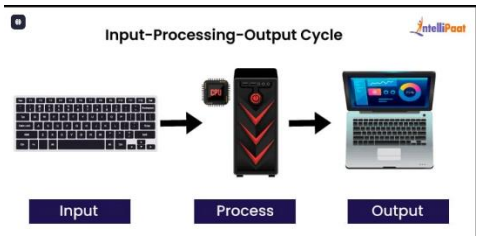
↓

Memory

↓

Storage Devices

1.2 Input Devices



Input devices are used to enter data and instructions into the computer.

The input unit is responsible for **entering data and instructions** into the computer.

Functions:

- Accepts data from the user
- Converts data into machine-readable (binary) form
- Transfers data to memory or CPU

Common input devices:

- **Keyboard** – Used for text and command input
- **Mouse** – Pointing and selection device
- **Scanner** – Converts hardcopy documents into digital form
- **Webcam** – Captures images and videos
- **Microphone** – Captures audio input

1.3 Output Devices

Output devices display or produce results after processing.

The output unit presents the **processed information** to the user in understandable form.

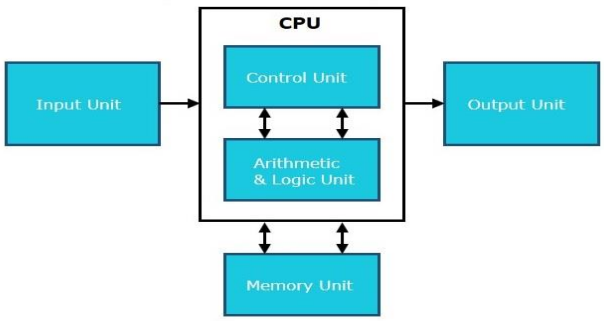
Functions:

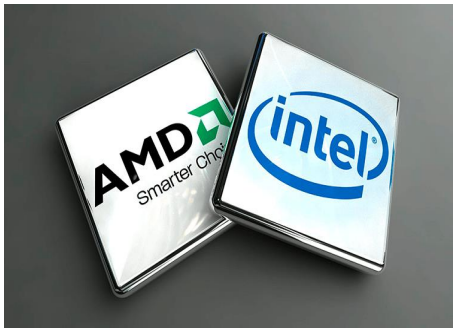
- Converts machine output into human-readable format
- Displays or produces results

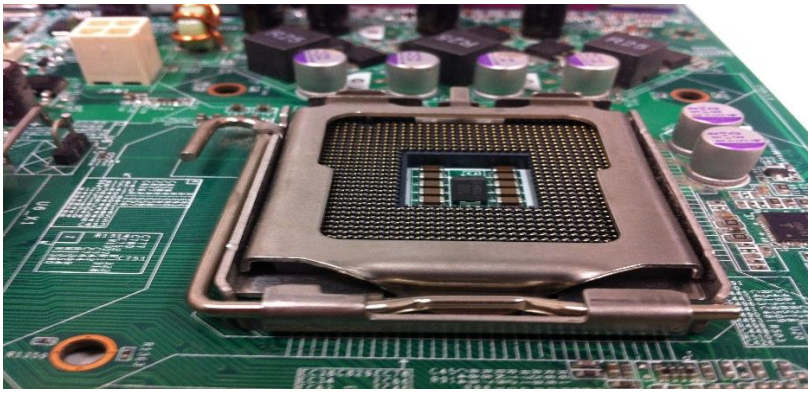
Examples:

- **Monitor** – Visual output
- **Printer** – Hardcopy output (inkjet, laser)
- **Speakers** – Audio output
- **Plotter** – Large-scale graphical output

Central Processing Unit (CPU)







CPU Introduction

The CPU is the **core component** of the computer system, often referred to as the *brain of the computer*.

Main functions:

- Executes program instructions
- Performs calculations and decision-making
- Controls and coordinates all system activities

Components of CPU:

- **Arithmetic Logic Unit (ALU)** – Performs arithmetic and logical operations
- **Control Unit (CU)** – Directs data flow and execution of instructions
- **Registers** – High-speed storage inside the CPU

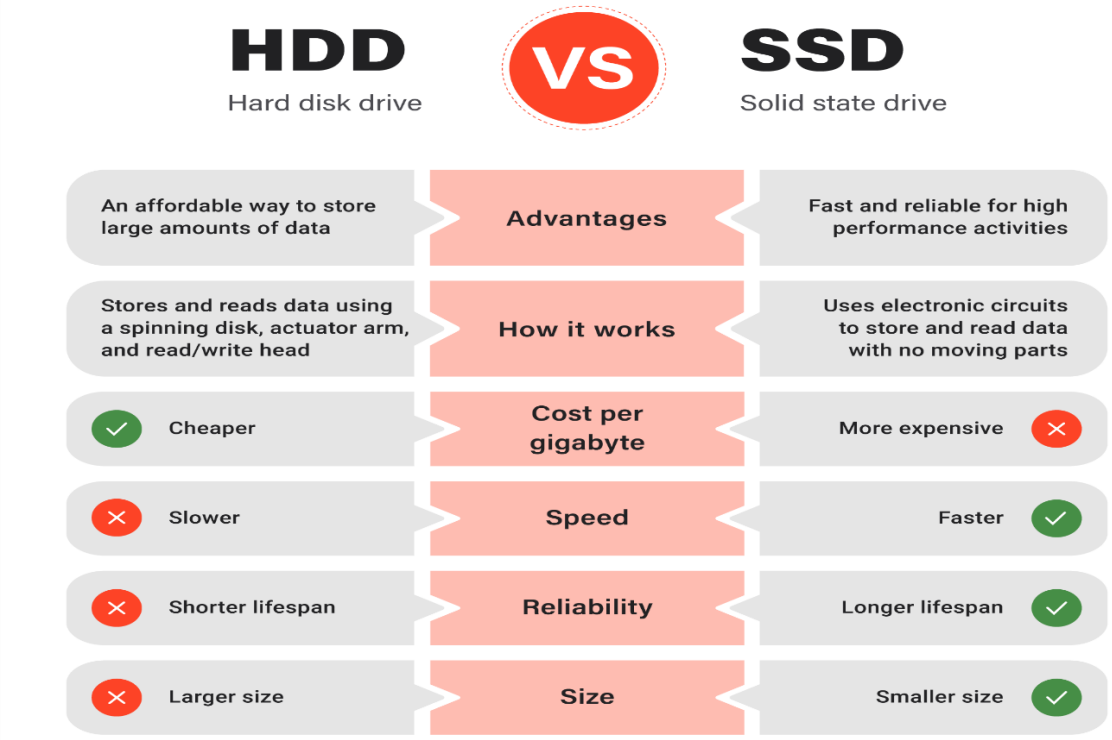
2.3 Processor Specifications

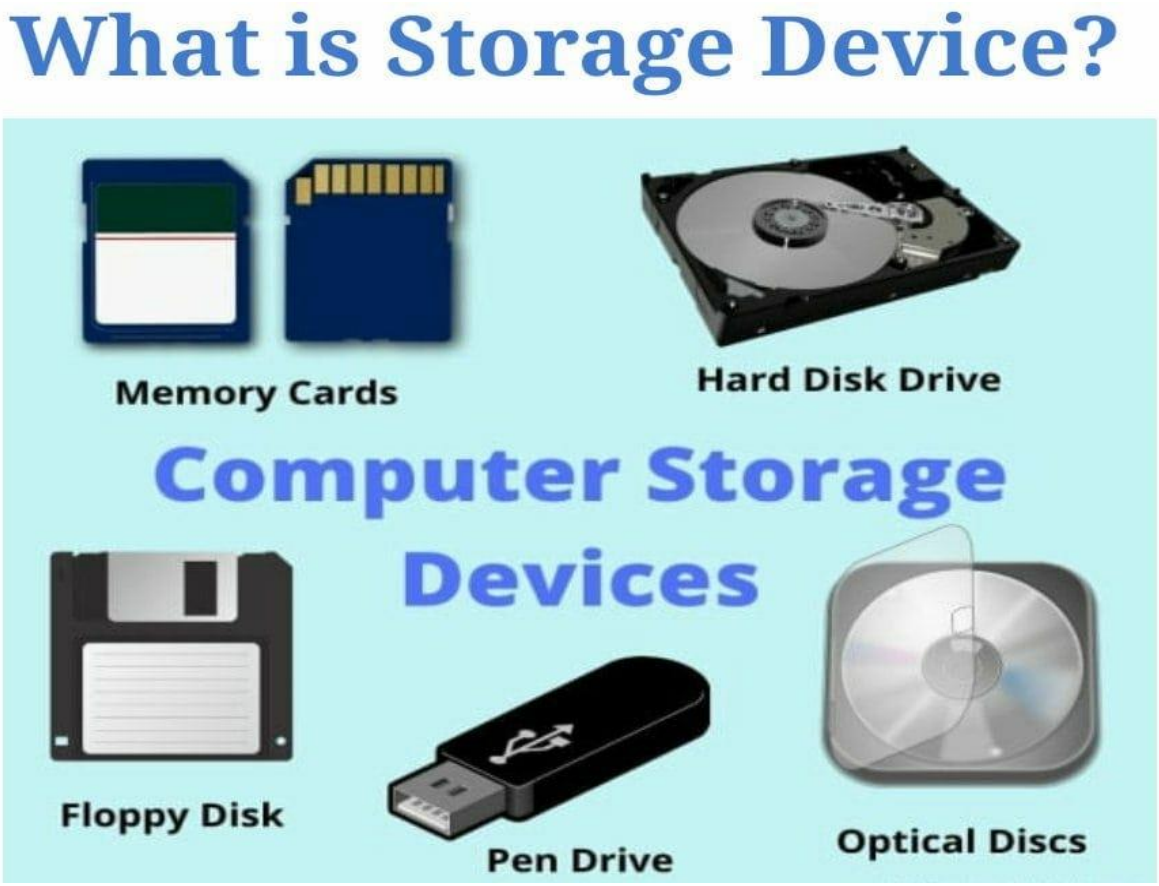
- **Clock Speed** – Measured in GHz
- **Cores** – Single-core, dual-core, quad-core, etc.
- **Threads** – Multiple instruction handling
- **Cache Memory** – L1, L2, L3 cache

Memory and Storage Devices

RAM	ROM
1. Temporary Storage.	1. Permanent storage.
2. Store data in MBs.	2. Store data in GBs.
3. Volatile.	3. Non-volatile.
4.Used in normal operations.	4. Used for startup process of computer.
5. Writing data is faster.	5. Writing data is slower.

Difference between RAM and ROM





3.1 Primary Memory

Primary memory is directly accessible by the CPU.

- **RAM (Random Access Memory)**
  Volatile, temporary storage
- **ROM (Read Only Memory)**
  Non-volatile, stores firmware

3.2 Secondary Storage Devices

Used for permanent data storage.

- **Hard Disk Drive (HDD)**
- **Solid State Drive (SSD)**
- **Optical Storage** – CD, DVD
- **Flash Storage** – Pen drive, memory card

Working of a Computer System (IPO Cycle)

1. **Input** – Data and instructions are entered using input devices
2. **Process** – CPU processes data using stored instructions
3. **Output** – Results are displayed through output devices
4. **Storage** – Data/results are saved for future use

This cycle continues until the task is completed.

Components Beyond Hardware

Software

- **System Software** – Operating systems, drivers
- **Application Software** – Word processors, browsers

Users

- People who interact with the computer system

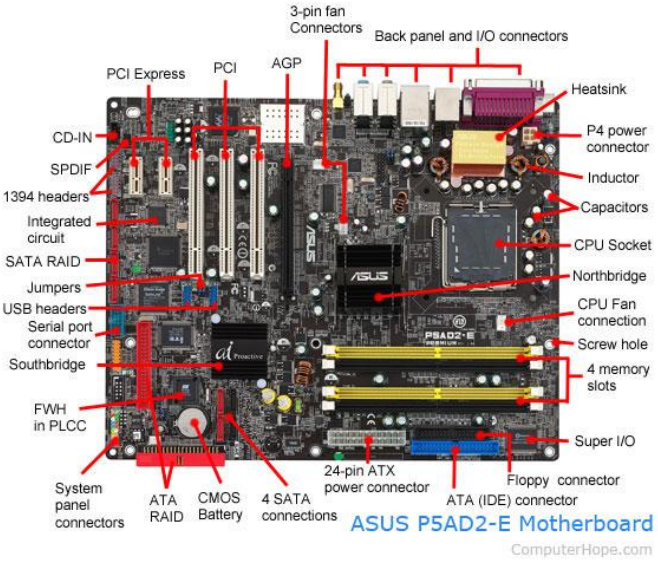
Procedures

- Rules and steps for operating the system

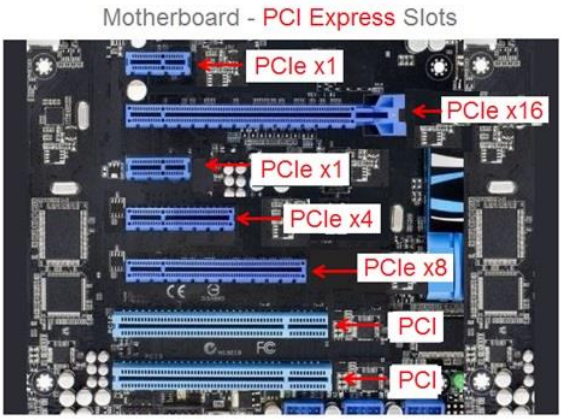
Importance of a Computer System

- High-speed data processing
- Accuracy and reliability
- Large storage capacity
- Automation of tasks
- Essential in education, healthcare, defense, business, and research

Motherboard and Expansion Cards







4.1 Motherboard

The motherboard is the **main circuit board** that connects all components.

Major parts:

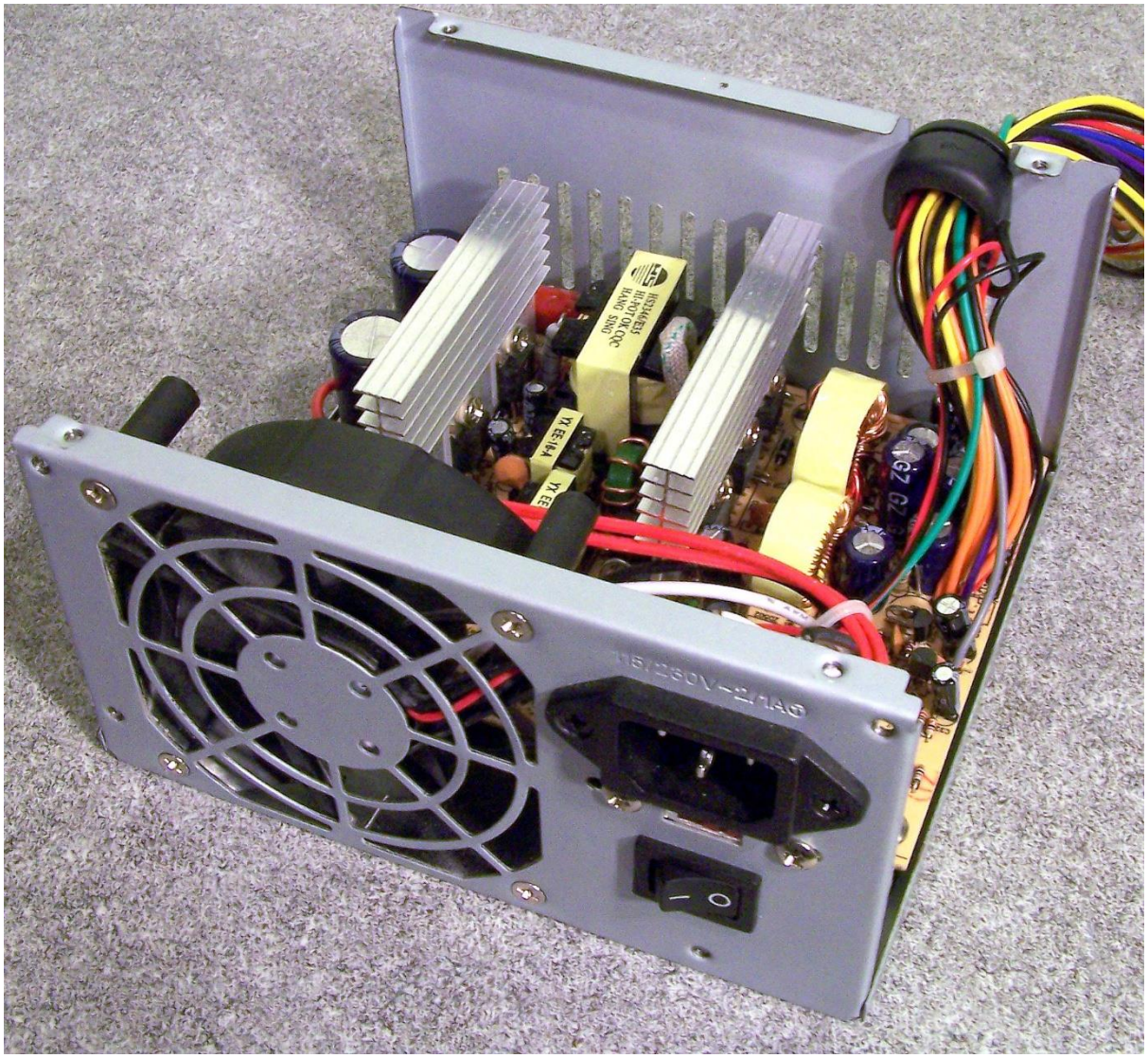
- CPU socket
- RAM slots
- Chipset
- BIOS/UEFI
- Expansion slots
- Power connectors

4.2 Expansion Cards

Used to extend system functionality.

- **Graphics Card**
- **Sound Card**
- **Network Interface Card (NIC)**

UNIT 5: Power Supply and Cooling System







5.1 Power Supply Unit (PSU)

The PSU converts AC power into DC power required by computer components.

Important features:

- Wattage rating
- Efficiency (80 PLUS)
- Connectors (24-pin, SATA, PCIe)

5.2 Cooling System

Prevents overheating of components.

- CPU fan
- Heat sink
- Case fans
- Liquid cooling (advanced systems)

UNIT 6: Computer Cabinet and Peripherals



Ultimate Chart of Computer Connectors and Ports

Updated for 2020

USB, Keyboard and Mouse


USB 2.0 Type A (male)


USB 2.0 Type A (female)


USB 2.0 Type B (male)


USB 2.0 Type B (female)


USB 2.0 Mini-A (male)


USB 2.0 Mini-B (male)


USB 2.0 Micro-A (male)


USB 2.0 Micro-B (male)


USB 3.0 Type A (male)


USB 3.0 Type A (female)


USB 3.0 Type B (male)


USB 3.0 Micro B (male)


USB 3.0 Type C (male)


USB 3.0 Type C (female)


PS/2 (male)


AT Keyboard (male)

Storage / Disk


SATA Type A


e-SATA


mSATA



M.2


U.2

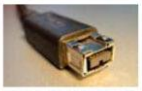

SAS (Serial Attached SCSI)

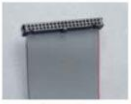

Lightning


Thunderbolt


Firewire 400 1394a 6-pin


Firewire 400 1394a 4-pin


Firewire 800/3200 1394b/c


IDE 40-pin


SCSI Micro DB68 (female)


SCSI Micro DB50 (male)


Internal 50-pin SCSI (f)


Internal 68-pin SCSI (f)

Network / Communications


GigE RJ-45 (SFP)


Ethernet RJ-45


Modem/Phone RJ-11


Serial/Parallel (f) DB-25


Serial RS232 (m) DE-9


Serial RS232 (f) DE-9


Centronics Parallel 36-pin


MIDI/gameport/DA-15

Audio


Line-Out, Headphones

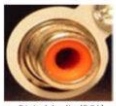

Microphone


Stereo Line In


Subwoofer/Center Out


Rear Surround Speakers


Mid Surround Speakers


Digital Audio (RCA)


Optical Audio "Toslink"

Video


VGA/SVGA (male) DE-15


VGA/SVGA (female) DE-15


DisplayPort (male)

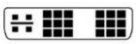

DisplayPort (female)


Mini DisplayPort (m)

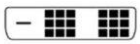

Mini DisplayPort (f)


HDMI Type A (male)


HDMI Type A (female)


DVI-I Single Link


DVI-I Dual Link


DVI-D Single Link


DVI-D Dual Link LFH-60


DVI-D Dual Link DMS-59


Micro-DVI


Mini-DVI


Mini HDMI Type C


Composite A/V (RCA)


S-Video


Component Video (m)


Component Video (f)


RF/Coaxial (male)


RF/Coaxial (female)


Apple Display Conn (ADC)


Micro HDMI Type D

Power


IEC 320 C13/C14 Connector


IEC 320 C13 Socket (f)


IEC 320 C14 Plug (m)


IEC 320 C19 (f)


IEC 320 C20 (m)


IEC 320 C5 Connector


SATA Power Connector


Molex 4-pin Connector

Brought to you by:

PRR Computers, LLC
(239) 244-1579
www.prrcomputers.com

This work is licensed under a Creative Commons Attribution-ShareAlike 4.0 International License.



6.1 Computer Cabinet

Houses all internal components.

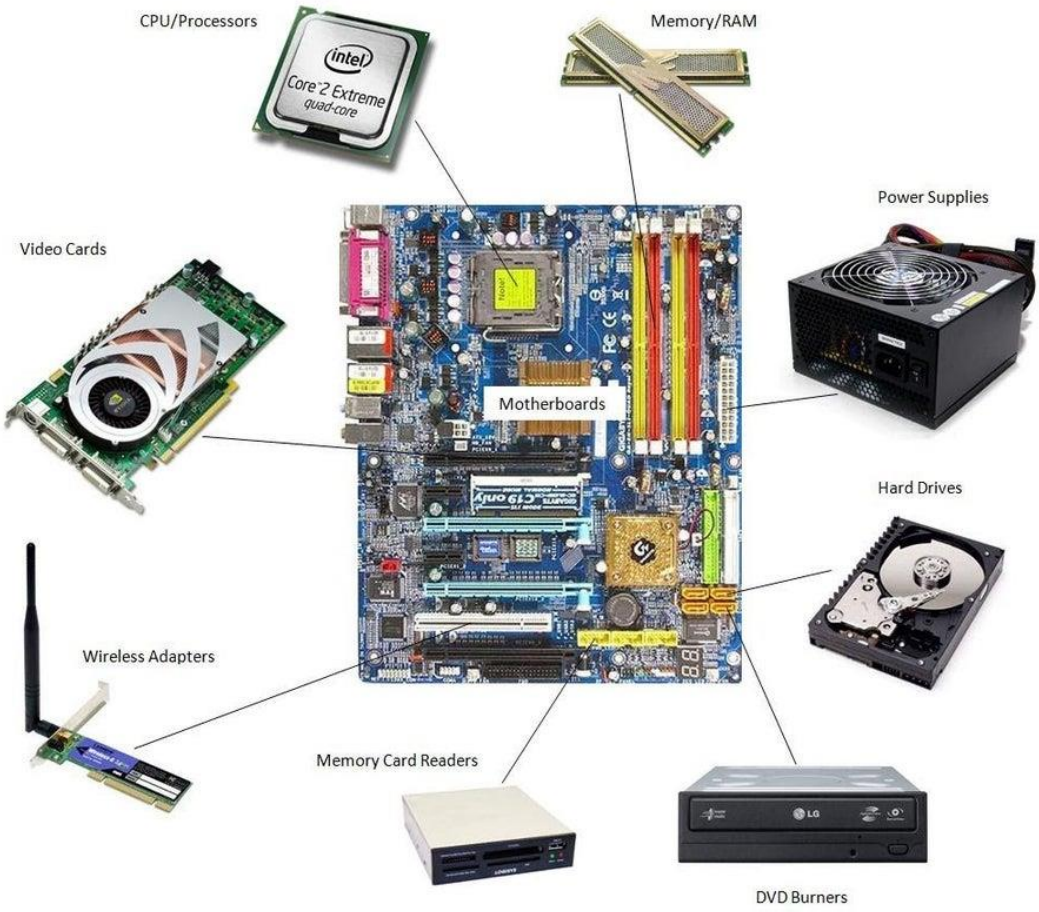
Types:

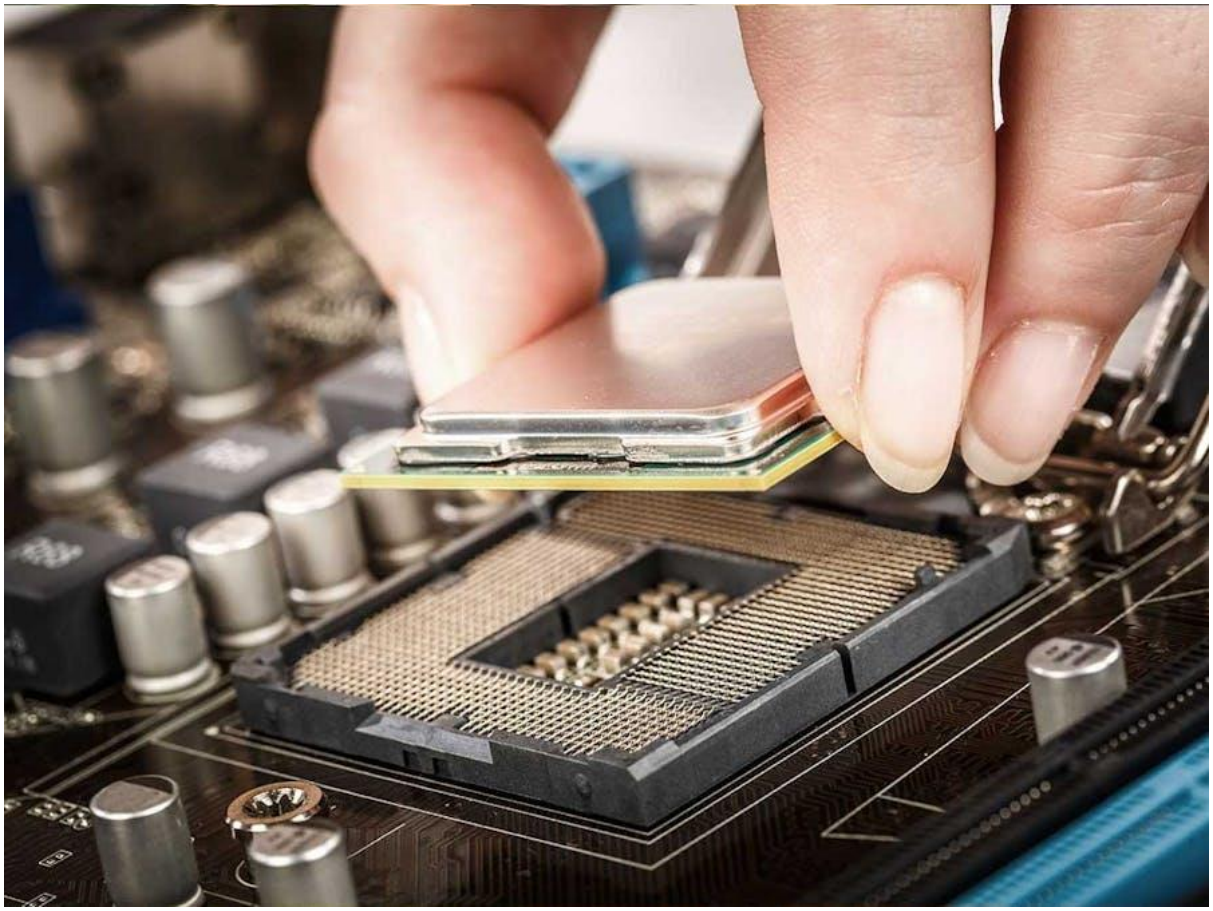
- Full Tower
- Mid Tower
- Mini Tower

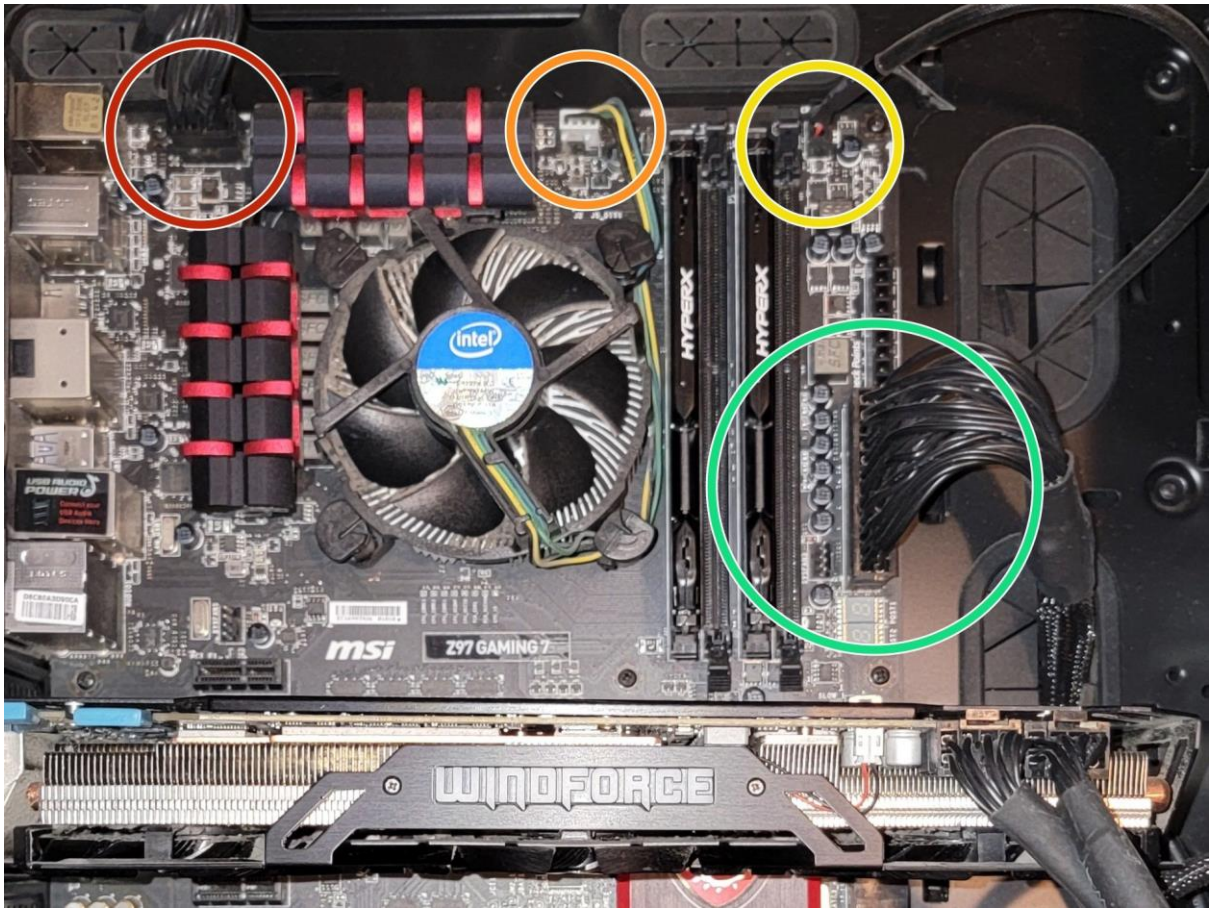
6.2 Ports and Connectors

- USB
- HDMI
- VGA
- Ethernet
- Audio ports

UNIT 7: Assembling a Computer System


CPU/Processors
Memory/RAM
Power Supplies
Video Cards
Motherboards
Hard Drives
Wireless Adapters
Memory Card Readers
DVD Burners




intel
msi
Z97 GAMING 7
HYPERX
HYPERX
WINDFORCE

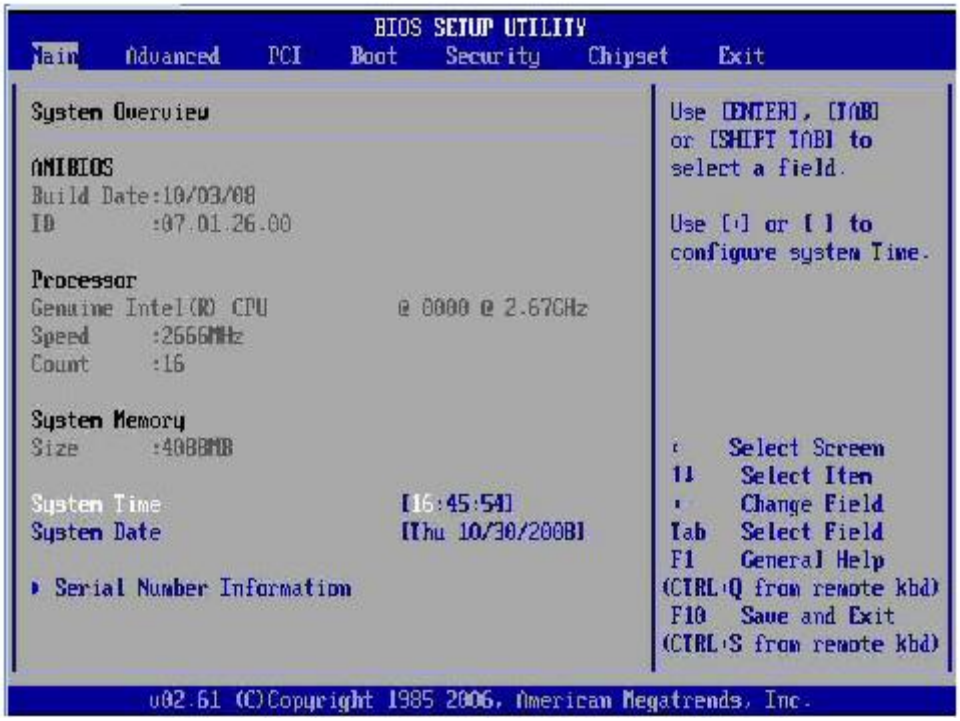
7.1 Pre-Assembly Precautions

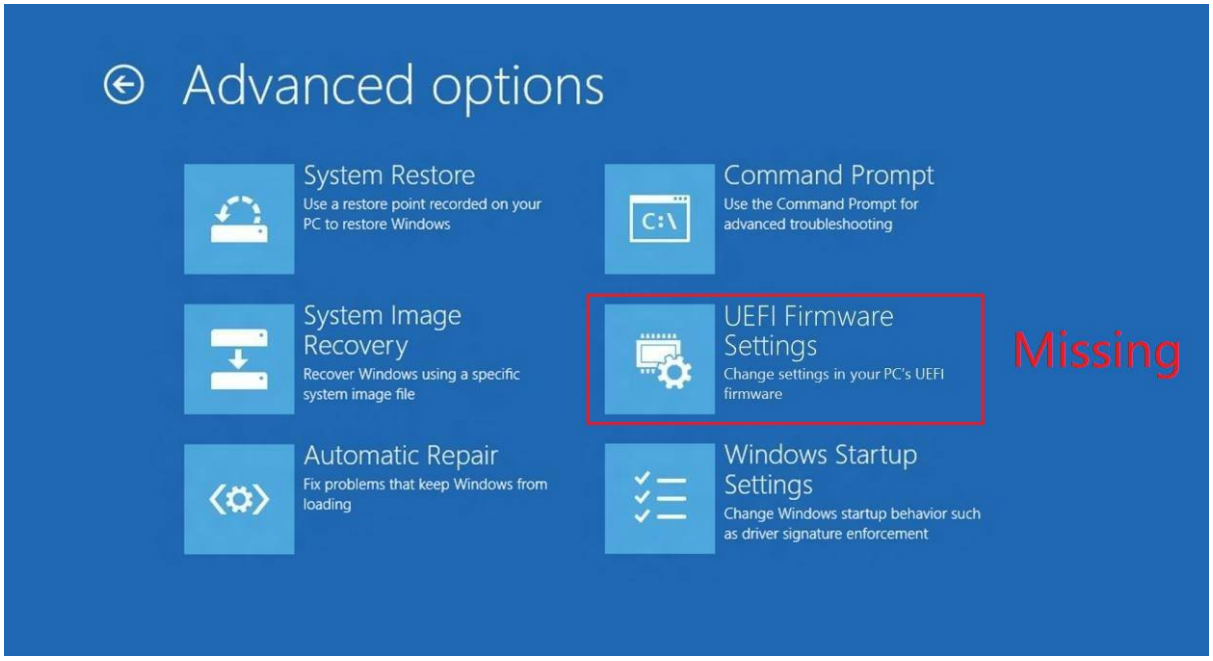
- Use anti-static wrist strap
- Work on a clean surface
- Handle components carefully

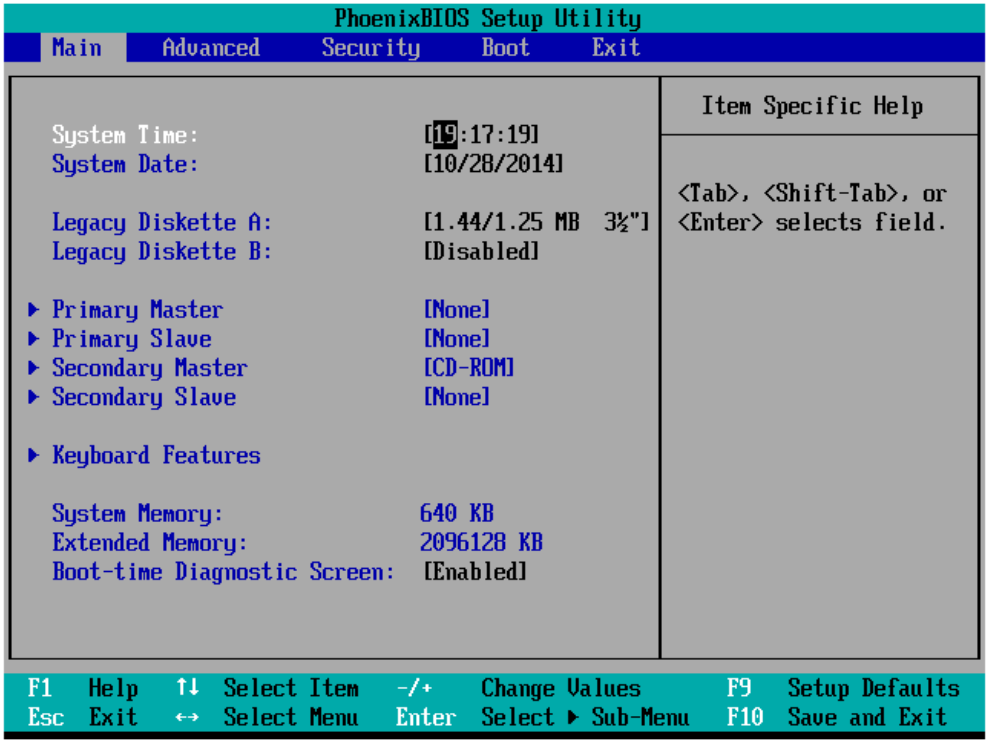
7.2 Assembly Steps

1. Install CPU on motherboard
2. Attach heat sink and fan
3. Insert RAM modules
4. Mount motherboard in cabinet
5. Install PSU
6. Connect storage devices
7. Attach expansion cards
8. Connect cables and peripherals

UNIT 8: BIOS/UEFI and System Configuration







8.1 BIOS/UEFI

Firmware that initializes hardware during booting.

Functions:

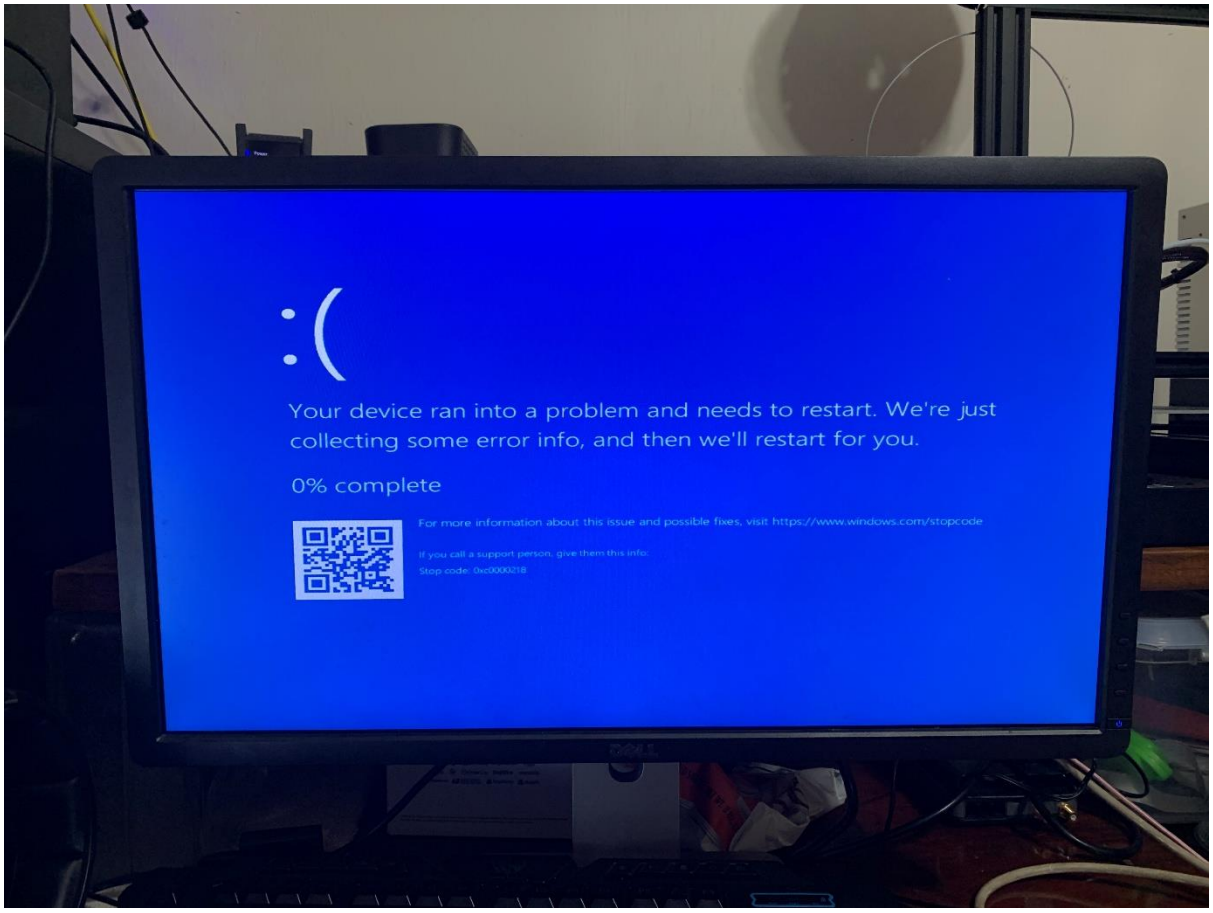
- Hardware detection
- Boot sequence setup

- System time and date
- Security settings

8.2 POST (Power On Self Test)

Checks hardware components before loading OS.

UNIT 9: Troubleshooting and Maintenance



9.1 Common Hardware Problems

- System not powering on
- No display
- Beep codes

- Overheating

9.2 Preventive Maintenance

- Regular cleaning
- Proper ventilation
- Software updates
- Antivirus protection

UNIT 10: Safety and E-Waste Management

10.1 Safety Measures

- Electrical safety
- Proper grounding
- Avoid liquid exposure

10.2 E-Waste Management

- Proper disposal of old hardware
- Recycling components
- Environmental responsibility
